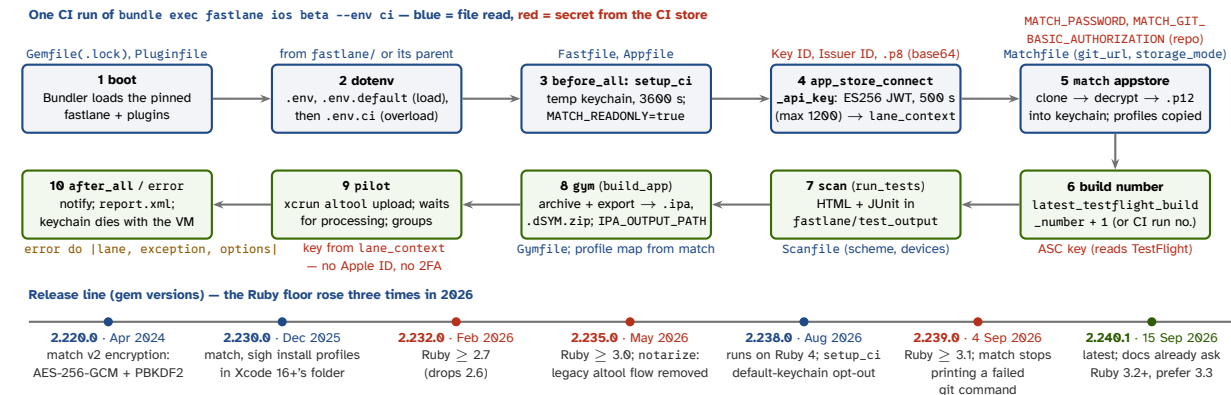


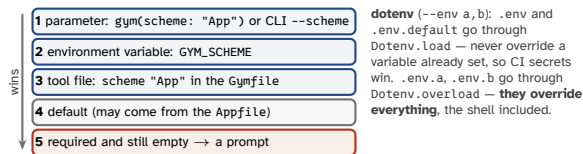
In one line: fastlane is a **Ruby gem** (pin it with **Bundler**) whose **lanes** call **actions**; gym, scan, pilot, deliver, snapshot, match, supply wrap xcodebuild, Apple's upload tools, the **App Store Connect API** and the **Google Play Developer API**. On CI its job is to make **signing and upload non-interactive**: match decrypts one shared cert/profile store into a **throw-away keychain**, and an **API key (.p8)** replaces an Apple ID that needs 2FA.



The files

Fastfile	lanes in Ruby; its code runs in <code>./fastlane</code> , actions + plugins in the project root
Appfile	<code>app_identifier</code> , <code>apple_id</code> (account e-mail), <code>team_id</code> , etc. <code>team_id</code> ; Android: <code>json_key_file</code> , <code>package_name</code>
Matchfile	per-tool defaults, found in <code>./fastlane</code> , <code>./fastlane</code> or <code>./fastlane</code> ; the first value set wins unless inside <code>for_lane / for_platform</code>
Gymfile ...	both committed; bundle <code>exec fastlane</code> ; upgrade with <code>bundle update fastlane</code>
Gemfile(.lock)	gem "fastlane-plugin-x" lines, pulled in by the Gemfile's <code>eval_gemfile</code>
Pluginfile	<code>README.md</code> (lane desc; FASTLANE_SKIP_DOCS), <code>report.xml</code> (JUnit of the actions)
generated	— <code>git-ignore</code> with <code>Preview.html</code> , <code>screenshots</code> , <code>test_output</code>

Where an option's value comes from



CLI `fastlane ios beta n:5 dry:yes` → `options[:n] == "5"`: only true/yes, false/no are converted; numbers stay strings.

Lanes

- platform :ios do → fastlane ios beta; desc feeds fastlane lanes and the README; a private_lane is hidden and callable only from another lane.
- A lane call is a method call: `build(release: true)`; last expression = return value; next leaves early and `after_each/after_all` still run.
- `before_all/after_all` once; `before_each/after_each` around every lane, called ones too; error on any raise.
- `lane_context[SharedValues::...]` chains actions: `upload_to_testflight` defaults ipa to `IPA_OUTPUT_PATH` and the key to `APP_STORE_CONNECT_API_KEY`.
- Plugins** are gems (`fastlane add_plugin x`) — third-party code on the signing machine: pin and review. `fastlane action gym` lists every option and its env name.

Tool ↔ action

`gym=build_app` · `scan=run_tests` · `pilot=upload_to_testflight` · `deliver=upload_to_app_store` · `snapshot=capture_ios_screenshots` · `precheck=check_app_store_metadata` · `match=sync_code_signing` · `cert=get_certificates` · `sigh=get_provisioning_profile` · `supply=upload_to_play_store`. **Deprecated:** `xcversion` → `xcodes`; `automatic_code_signing` → `update_code_signing_settings`. **precheck** runs inside deliver before submit at level warn: placeholder text, other platforms, unreachable URLs.

Example — Fastfile

```
before_all { setup_ci } # no-op unless a CI env var is set
platform :ios do
  lane :beta do
    app_store_connect_api_key(key_id: ENV["ASC_KEY_ID"],
                              issuer_id: ENV["ASC_ISSUER_ID"], key_content: ENV["ASC_P8_B64"],
                              is_key_content_base64: true) # -> lane_context
    match(type: "appstore", readonly: true)
    increment_build_number(build_number: latest_testflight_build_number + 1)
    build_app(scheme: "App", export_method: "app-store")
    upload_to_testflight(skip_waiting_for_build_processing: true)
  end
end
```

`increment_build_number` agvtool new-version -all <n>; with no n, next-version — which needs Apple Generic versioning. An empty train returns `initial_build_number` (1).

match — storage and encryption

git (default)	encrypted with MATCH_PASSWORD; branch master, one branch per team; auth
s3	git_basic_authorization (base64), _bearer..., git_private_key encrypted unless s3_skip_encryption
google_cloud	not encrypted by match — Google-managed keys; bucket IAM is the lock
gitlab_secure_files	not encrypted by match — GitLab project access is the lock
v1	Salted... AES-256-CBC · key + IV from one MD5 round (SHA-256 retried) — a wrong password went undetected ~0.4% of the time
v2 (2.220.0+)	match_encrypted_v2... AES-256-GCM + tag: PBKDF2-HMAC-SHA256, 10 000 rounds, 8-byte salt. Older fastlane cannot read it: force_legacy_encryption. The docs' openssl aes-256-cbc recipe opens v1 only; fastlane match decrypt opens both

match — what it writes and installs

- `certs/<type>/<cert-id>.cer` + `.p12` (distribution serves appstore and adhoc) · `profiles/appstore/AppStore-<bundle>.mobileprovision`; portal name `match AppStore <bundle>`.
- security `import -T /usr/bin/codesign` into `keychain_name` (default `login.keychain`, on CI `setup_ci`'s), then `set-key-partition-list -S apple-tool:,apple:,codesign:` — signing without a UI prompt.
- Profiles go to `~/Library/Developer/Xcode/UserData/Provisioning Profiles` for Xcode ≥ 16 (`~/Library/MobileDevice/...` before; right only since 2.230.0) — select Xcode (xcodes) before match.
- Sets `ENV["sigh-<bundle>.appstore"]` = profile UUID (+ `_profile-name`, `_team-id` ...), `MATCH_PROVISIONING_PROFILE_MAPPING` for gym, `SIGH_PROFILE_TYPE` as gym's default `export_method`.
- readonly fetches only · `force_for_new_devices` renews when the portal's device count changed · `change_password` · `import` (existing `.cer/.p12`) · `migrate` (storage → storage) · `nuke` distribution revokes + deletes: App Store / TestFlight builds keep working, **Ad Hoc and Enterprise builds stop**.

Auth — API key vs Apple ID

ASC API key	Key ID + Issuer ID + .p8 (downloadable once). Team key: an Admin makes it, role-scoped, all apps. Individual key: no Provisioning, Sales and Finance or notaryTool — match, sigh, cert need a team key. Apple rejects tokens > 20 min for most requests (scoped GET-only tokens on a few resources: up to 6 months). Works for pilot, deliver, match, sigh, cert, precheck (not IAP); not pem
Apple ID	FASTLANE_USER + password + 2FA; fastlane spaceauth → FASTLANE_SESSION, valid 1 day - 1 month and only in the region it was made — make it on the CI machine. An app-specific password covers binary upload only . It drives Apple's web sign-in: 2.240.0 had to re-find the sign-in widget key after Apple dropped its endpoint

supply — Google Play

- Service-account JSON (`json_key / _data`). One run = one Play Developer API **edit**: insert → upload AAB → track → listing → commit (or validate_only). **A commit or a Play Console change invalidates every other open edit** — serialise jobs.
- track production (default), beta, alpha, internal; `release_status` completed (default), draft, halted, inProgress; `rollout: 0.1` → `inProgress` + `userFraction`, `rollout: 1` → completed.

Traps

- `sh("ls")` runs in `./fastlane`, actions in the root — relative paths differ.
- No `bundle exec` = the global fastlane; plugins may not load.
- A committed `.env.ci` overrides the CI secret of the same name.
- `-showBuildSettings` times out on big projects (3s × 3): `FASTLANE_XCODEBUILD_SETTINGS_TIMEOUT`.
- `export_method` takes only app-store, ad-hoc ...; Xcode 15.4+ logs them as deprecated (app-store-connect, release-testing).
- `pilot's apple_id` is the app's numeric id, not the Appfile's e-mail; `skip_waiting_for_build_processing` → nothing is distributed.

Related: Code signing & iOS CI/CD · macOS keychains & the security CLI (CI signing) · Signing failures — symptom → check → fix · React Native CI/CD & Android signing — build once, promote · Supply-chain security — from a maintainer's laptop to your app · xcodebuild-simctl-cli · github-actions-deep · ci-pipeline-design